Max Chung

Local formulas

$$G(z) = E(z^X) = \sum_{k=0}^{\infty} z^k p(k)$$

$$G^{(k)}(0) = k!\, p(k)$$

$$\Rightarrow p(k) = \frac{1}{k!} G^{(k)}(0)$$

$$\sigma^2 = Var(X) = G''(1) + G'(1) - (G'(1))^2$$

$$\hookrightarrow G''(1) + E(X) - (E(X))^2$$

$$M(t) = E(e^{tX})$$

$$M^{(k)}(0) = E(X^k)$$

$$h(x) = \frac{f(x)}{S(x)} = -\frac{d}{dx} \ln(S(x))$$

$$\Rightarrow S(x) = e^{-\int_0^x h(y)\,dy}$$

$$f_X(x) = \int_{\mathbb{R}} f(x|y) f(y)\,dy$$

$$Var(X+Y) = Var(X) + Var(Y) + 2\,Cov(X,Y)$$

$$Cov(X,Y) = E(XY) - E(X)E(Y)$$

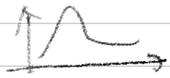
$$E(X|Y) = \int x \cdot f(x|y)\,dx$$

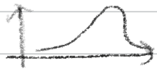
$$Var(X|Y) = E(X^2|Y) - (E(X|Y))^2$$

$CV(X) = \frac{\sigma}{\mu}$

$Skewness(X) = \frac{E[(X-\mu_x)^3]}{\sigma^3_x}$

$=0$, Normal

>0, right skewed 

<0, left skewed 

$Kurt(X) = \frac{E[(X-\mu_x)^4]}{\sigma^4_x}$

$=3$, Normal

>3, peaked

<3, flat.

$$\left.\begin{array}{l}(X-d)^+ = \max\{0, X-d\} \\ X\wedge d = \min\{d, X\}\end{array}\right) + = X$$

$E((X-d)^+) = \int_d^\infty x f(x)\,dx - d\cdot S_x(d)$

$E(X\wedge d) = \int_{-\infty}^d x f(x)\,dx + d\cdot S_x(d)$

$E[((X-d)^+)^k] = \int_d^\infty k(x-d)^{k-1} S_x(x)\,dx$

$E[(X\wedge d)^k] = \int_0^d kx^{k-1} S_x(x)\,dx$

π_p = pth quantile.

Monetary RM (ρ) :

1. $\rho(X+c) = \rho(X) + c$
2. $\rho(X_1) \le \rho(X_2) \Leftrightarrow X_1 \le X_2$,

↳ Convex RM : $\rho(\lambda X_1 + (1-\lambda)X_2) \le \lambda\rho(X_1) + (1-\lambda)\rho(X_2)$

↳ Coherent RM : $\rho(\lambda X) = \lambda\rho(X)$, $\lambda > 0$

$\Rightarrow \rho(X_1 + X_2) \le \rho(X_1) + \rho(X_2)$

$\mathrm{VaR}_p(X) = \inf\{x \in \mathbb{R} : \Pr(X > x) \le 1-p\}$

cts $\Rightarrow$ sol. to $\Pr(X > \mathrm{VaR}_p(X)) = 1-p$.

basically 100pth percentile of X

$x \ge \mathrm{VaR}_p(X) \Leftrightarrow S_X(x) \le 1-p$

$\mathrm{TVaR}_p(X) = \frac{1}{1-p}\int_p^1 \mathrm{VaR}_q(X)\,dq$

$\mathrm{TVaR}_p(X) \ge \mathrm{VaR}_p(X)$

Premium = $\underbrace{\text{loss} + \text{Expense}}_{\text{Cost.}}$ + Profit.

loss $= X$, premium $= P = \pi(X)$, $\theta > 0 \Rightarrow$ risk loading.

Pure premium : $\pi(X) = E(X)$

Expected Value premium : $\pi(X) = (1+\theta)E(X)$

Mean - S.D. premium : $\pi(X) = E(X) + \theta\sqrt{\mathrm{Var}(X)}$

Mean-Variance premium : $\pi(X) = E(X) + \theta\,\mathrm{Var}(X)$

Dutch premium : $\pi(X) = E(X) + \theta E((X - E(X))^+)$

Properties of Premium :

1. Non-negative risk loading $\pi(X) \ge E(X)$
2. Indep. additivity : $\pi(X_1 + X_2) = \pi(X_1) + \pi(X_2)$, X_1, X_2 indep.
3. Scale invariance : $\pi(cX) = c\pi(X)$
4. Cash invariance : $\pi(c + X) = c + \pi(X)$
5. No rip-off : $\pi(X) \le \max\{X\}$

N # claim w/ pmt amount X_i

$S = \sum_{i=1}^{N} X_i$, $N > 0$

$X \sim Uni(a, b)$:

$f(x) = \frac{1}{b-a}$, $F(x) = \frac{x-a}{b-a}$, $S(x) = \frac{b-x}{b-a}$

$E(X) = \frac{b+a}{2}$, $Var(X) = \frac{(b-a)^2}{12}$

$X \sim GAM(\alpha, \theta)$

$f(x) = \frac{x^{\alpha-1} e^{-x/\theta}}{\Gamma(\alpha)\theta^{\alpha}}$, $E(X) = \alpha\theta$, $Var(X) = \alpha\theta^2$

$\Gamma(k) = (k-1)!$

$X \sim Exp(\theta)$

$f(x) = \frac{1}{\theta} e^{-\frac{1}{\theta}x}$, $F(X) = 1 - e^{-\frac{1}{\theta}x}$, $S(x) = e^{-\frac{1}{\theta}x}$

$E(X) = \theta$, $Var(X) = \theta^2$

$X \sim PAR(\alpha, \theta)$

$f(x) = \frac{\alpha\theta^{\alpha}}{(x+\theta)^{\alpha+1}}$, $x > 0$ $F(x) = 1 - \left(\frac{\theta}{x+\theta}\right)^{\alpha}$, $S(x) = \left(\frac{\theta}{x+\theta}\right)^{\alpha}$

$E(X) = \frac{\theta}{\alpha-1}$, $\alpha > 1$ $Var(X) = \frac{\alpha\theta^2}{(\alpha-1)^2(\alpha-2)}$, $\alpha > 2$.

mgf DNE.

$X \sim WEL(\theta, \tau)$ $(Exp = WEL(\theta, 1))$

$f(x) = \frac{\tau\left(\frac{x}{\theta}\right)^{\tau} e^{-\left(\frac{x}{\theta}\right)^{\tau}}}{x}$, $x > 0$ $F(x) = 1 - e^{-\left(\frac{x}{\theta}\right)^{\tau}}$, $S(x) = e^{-\left(\frac{x}{\theta}\right)^{\tau}}$

$E(X) = \theta\,\Gamma(1 + \frac{1}{\tau})$, $Var(X) = \theta^2\left[\Gamma(1+\frac{2}{\tau}) - \left(\Gamma(1+\frac{1}{\tau})\right)^2\right]$

$X \sim N(\mu, \sigma^2)$

$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$, $F(X) = Z\left(\frac{x-\mu}{\sigma}\right)$, $S(X) = 1 - Z\left(\frac{x-\mu}{\sigma}\right)$

Non-parametric: (constructing distribution)

Assume $\{X_1, \dots X_n\}$,

$$F(X) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{I}\{X_i \le X\} \longrightarrow \mathbb{I}\{X_i \le X\} = \begin{cases} 1 & \text{true} \\ 0 & \text{false.} \end{cases}$$

Parametric: all distribution above are parametric.

Transforming RVs.

- $Y = cX$: (scale distribution) p.s. X cts.

$$F_Y(y) = P(cX \le y) = P(X \le \tfrac{y}{c}) = F_X(\tfrac{y}{c})$$

$$f_Y(y) = \frac{d}{dy}F_Y(y) = \frac{1}{c} f_X(\tfrac{y}{c})$$

- $Y = X^{\frac{1}{c}}$ $\quad X \ge 0$ cts.

$$F_Y(y) = P(X^{\frac{1}{c}} \le y) \overset{c>0}{=} F(y^c) \Rightarrow f_Y(y) = c y^{c-1} f_X(y^c)$$

$$\overset{c<0}{=} 1 - F(y^c) \Rightarrow f_Y(y) = -c y^{c-1} f_X(y^c)$$

- $Y = e^X$ $\quad X \ge 0$ cts.

$$F_Y(y) = P(e^X \le y) = P(X \le \ln(y)) = F_X(\ln(y))$$

$$f_Y(y) = \frac{1}{y} f_X(\ln(y))$$

- $T_t = X - t \mid X > t$ ☆ (residual lifetime RV) p.s. $X \ge 0$ cts., $t > 0$.

$$S_{T_t}(x) = \frac{S_X(x+t)}{S_X(t)} \qquad F_{T_t}(x) = \frac{P(t < X \le x+t)}{S_X(t)}$$

$$f_{T_t}(x) = \frac{f_X(x+t)}{S_X(t)}$$

$$E(X) = \frac{1}{S_X(t)}\int_t^\infty S_X(x)\,dx \triangleq e_X(t)$$

$$\lim_{t\to\infty} e_X(t) = \lim_{t\to\infty} \frac{-S_X(t)}{-f_X(t)} = \lim_{t\to\infty} \frac{1}{h_X(t)}$$

Mixture of Distribution:

Θ = discrete / cts. mixing RV.

- Discrete $\Theta = \{\theta_1, \theta_2 \dots \theta_n\}$

  w/ pmf $f_\Theta(\theta_i) = P(\Theta = \theta_i) > 0$, $i = 1 \dots n$

  $F_{X|\Theta}(x|\theta_i) = P(X \le x_i \mid \Theta = \theta_i)$

  $\Rightarrow F_X(x) = \sum_{i=1}^{n} P(X \le x \mid \Theta = \theta_i) P(\Theta = \theta_i)$

  $= \sum_{i=1}^{n} F_{X|\Theta}(x|\theta_i) f_\Theta(\theta_i)$

- Cts Θ

  $F_{X|\Theta}(x|\theta) = P(X \le x_i \mid \Theta = \theta)$

  $\Rightarrow F_X(x) = \int_{-\infty}^{\infty} P(X \le x \mid \Theta = \theta) f_\Theta(\theta)\, d\theta$

  $= \int_{-\infty}^{\infty} F_{X|\Theta}(x|\theta) f_\Theta(\theta)\, d\theta$

Theorem: $E(h(X)) = E(E(X|\Theta))$

$Var(X) = E(Var(X|\Theta)) + Var(E(X|\Theta))$

Measure of tail:

- Criterion 1: existence of all moments $\Rightarrow$ light-tailed

  $= E(X^k)$? existence ... up to certain k $\Rightarrow$ heavy-tailed

- Criterion 2:

  limiting ratio $c := \lim_{x\to\infty} \frac{S_X(x)}{S_Y(x)} = \lim_{x\to\infty} \frac{f_X(x)}{f_Y(x)}$

  $c = 0 \Rightarrow Y$ heavier tail than X

  $0 < c < \infty \Rightarrow X$ and Y equivalent tail

  $c = \infty \Rightarrow X$ heavier tail than Y

- Criterion 3: for large x

$h(x)$ decreasing $\Rightarrow$ X heavy-tailed. (DFR)

$h(x)$ increasing $\Rightarrow$ X light tailed. (IFR)

D/IFR = Decreasing / Increasing failure rate.

note: exponential is both DFR and IFR

- Criterion 4:

$$e_X(d) = \frac{\int_d^\infty S_X(x)\,dx}{S_X(d)}$$

$e_X(d)$ increasing $\Rightarrow$ IMRL (decreasing mean residual lifetime)

$e_X(d)$ decreasing $\Rightarrow$ DMRL (increasing ...)

note: DMRL = IFR $\Rightarrow$ light-tailed.

IMRL = DFR $\Rightarrow$ heavy-tailed

Policy adjustments.

$I(X)$ = insurer pmt. amount.

↳ increasing, indemnification function.

Order: policy limit → policy deductible → coinsurance.

Reporting Methods:

x ① Loss basis: Y_L = amount paid per loss

$F_{Y_L}(y) = P(Y_L \le y) = P(I(X) \le y)$

v ② Payment basis: Y_P = amount paid per pmt.

$$F_{Y_P}(y) = P(Y_L \le y \mid Y_L > 0) = \frac{F_{Y_L}(y) - F_{Y_L}(0)}{1 - F_{Y_L}(0)}$$

no deductible $\Rightarrow F_{Y_P}(y) = F_{Y_L}(y)$ (since $F_{Y_L}(0) = 0$)

deductible $\Rightarrow F_{Y_L}(0) = P(X \le d)$

- Policy limits:

$I(X) = \min\{X, u\} = X \wedge u$

u = maximum loss level / maximum pmt.

$\nwarrow$ aka term policy limit.

w/ only policy limit, $Y_L = Y_P = I(X) = X \wedge u$

$f_Y(y) = f_X(y)$

$P(Y = u) = S_X(u) = P(X > u)$

$$F_Y(y) = \begin{cases} F_X(y), & 0 \leq y < u \\ 1, & y \geq u \end{cases}$$

$$S_Y(y) = \begin{cases} S_X(y), & 0 \leq y < u \\ 0, & y \geq u \end{cases}$$

$E(Y) = E(X \wedge u) = \int_0^u y f_X(y)\, dy + u S_X(u) = \int_0^u S_X(y)\, dy$

$Var(Y) = \int_0^u 2y\, S_X(y)\, dy - \left(\int_0^u S_X(y)\, dy\right)^2$

- Ordinary deductible:

$I(X) = \max\{X - d, 0\} = (X - d)^+$

$Y_L = I(X) = (X - d)^+$ (mixed)

$P(Y_L = 0) = P(X \leq d) = F_X(d)$

$f_{Y_L}(y) = f_X(y + d)$

$F_{Y_L}(y) = F_X(y + d)$

$E(Y_L) = \int_d^\infty (y - d) f_X(y)\, dy = \int_d^\infty S_X(y)\, dy$.

$Var(Y_L) = \int_d^\infty 2(y - d)\, S_X(y)\, dy - \left(\int_d^\infty S_X(y)\, dy\right)^2$

$Y_P = I(X) = (Y_L \mid Y_L > 0) = (X-d \mid X > d)$ (cts)

$$f_{Y_P}(y) = \frac{f_{Y_L}(y)}{1-F_{Y_L}(0)} = \frac{f_X(y+d)}{1-F_X(d)}$$

$$F_{Y_P}(y) = \frac{F_{Y_L}(y)-F_{Y_L}(0)}{1-F_{Y_L}(0)} = \frac{F_X(y+d)-F_X(d)}{1-F_X(d)}$$

$$S_{Y_P}(y) = \frac{S_{Y_L}(y)}{S_{Y_L}(0)} = \frac{S_X(y+d)}{S_X(d)}$$

$$E(Y_P) = e_{Y_L}(0) = e_X(d) = \frac{1}{S_X(d)}\int_d^\infty S_X(y)\,dy = \frac{E(Y_L)}{S_X(d)}$$

$$Var(Y_P) = \frac{E(Y_L^2)}{S_X(d)} - \left(\frac{E(Y_L)}{S_X(d)}\right)^2$$

- Franchise deductible d

$$I(X) = X \cdot \mathbb{I}\{X > d\} \quad (0 \text{ or } X)$$

- Coinsurance factor α $(0 \le \alpha \le 1)$

$$Y_P = Y_L = I(X) = \alpha X$$

$$f_Y = \frac{1}{\alpha} f_X\left(\frac{y}{\alpha}\right) \quad \text{(multiply distribution)}$$

$$F_Y(y) = F_X\left(\frac{y}{\alpha}\right)$$

$$S_Y(y) = S_X\left(\frac{y}{\alpha}\right)$$

$$E(Y) = \alpha E(X)$$

$$Var(Y) = \alpha^2 E(X)$$

Multiple Modifications:

follow order: limit → deductible → coinsurance.

assume all above, with:

limit u, deductible d, coinsurance α.

↓

$$X \xrightarrow{u} Y_L = Y_P = X \wedge u$$

$$\xrightarrow{d} \begin{cases} Y_L = (X\wedge u - d)^+ \\ Y_P = \underline{X\wedge u} - d \mid X > d \end{cases}$$

$$\xrightarrow{\alpha} \begin{cases} Y_L = \alpha(X\wedge u - d)^+ \\ Y_P = \alpha(\underline{X\wedge u} - d) \mid X > d \end{cases}$$

u = maximum loss covered

$\alpha(u-d)$ = maximum payment amount

let $Y_L = \alpha(X\wedge u - d)^+$, then

limit
→ deductible
→ coinsurance.

$$S_{Y_L}(y) = \begin{cases} 1 & , y < 0 \\ S_X(\frac{y}{\alpha} + d) & , \underline{0 \le y < \alpha(u-d)} \\ 0 & , y \ge \alpha(u-d) \end{cases}$$

$$S_{Y_P}(y) = \begin{cases} 1 & , y < 0 \\ \dfrac{S_X(\frac{y}{\alpha} + d)}{S_X(d)} & , \underline{0 \le y < \alpha(u-d)} \\ 0 & , y \ge \alpha(u-d) \end{cases}$$

loss elimination ratio:

def: proportion of loss eliminate during modification.

$$LER = 1 - \frac{E(Y_L)}{E(X)}$$

for $u > 0$, $LER = \dfrac{\int_u^\infty S_X(y)\,dy}{E(X)} = \dfrac{E[(X-u)^+]}{E(X)}$

for $d > 0$, $LER = 1 - \dfrac{\int_d^\infty S_X(t)\,dy}{E(X)} = \dfrac{E(X\wedge d)}{E(X)}$

Counting RVs.

$p_k = P(N=k)$

$G_N(t) = E(t^X) = \sum_{k=0}^{\infty} t^k p_k$

$M_N(t) = E(e^{tX}) = \sum e^{tk} p_k$

$G_N(t) = M_N(\ln(t))$

☆ $p_0 = G_N(0)$ ☆ $p_n = \frac{1}{n!} G_N^{(n)}(0)$

☆ $G_N^{(n)}(1) = E(N(N-1) \cdots (N-n+1))$

☆ $Var = G''(1) + E(X) - (E(X))^2$

$N \sim Poi(\lambda)$

$p_k = \frac{\lambda^k e^{-\lambda}}{k!}$

$G_N(t) = e^{\lambda(t-1)}$

$N_1 + \cdots N_m = Poi(\lambda_1 + \cdots \lambda_m)$

compound.

$N_i \sim Poi(\lambda p_i)$, $i = 1, 2 \ldots k$, iid.

$N \sim Bin(m, q)$

$\binom{m}{k} = \frac{m!}{k!(m-k)!}$

$p_k = \binom{m}{k} q^k (1-q)^{m-k}$

$G_N(t) = (1-q-tq)^m$

$E(N) = mq$, $Var(N) = mq(1-q)$

let $N_m \sim Bin(m, \frac{\lambda}{m})$

$\Rightarrow N_m \to Poi(\lambda)$ as $m \to \infty$

$N_1 + \cdots N_n = Bin(m_1 + \cdots m_n, q)$

$\quad\quad\quad\quad p = \frac{1}{1+\beta}$

$N \sim NB(\beta, r)$

$p_k = \binom{k+r-1}{k}\left(\frac{1}{1+\beta}\right)^r\left(\frac{\beta}{\beta+1}\right)^k$

$G_N(t) = (1+\beta-\beta t)^{-r}$

$E(N) = r\beta$, $Var(N) = r\beta(1+\beta)$

for $r=1$, $p_k = \frac{1}{1+\beta}\left(\frac{\beta}{1+\beta}\right)^t$

$F_N(k) = 1-\left(\frac{\beta}{1+\beta}\right)^{k+1}$

$G_N(t) = (1+\beta-\beta t)^{-1}$

$N_1 + \dots N_m \sim \underline{NB(\beta, r_1 + \dots r_m)}$

$(a,b,0) =$

$p_k = \left(a+\frac{b}{k}\right)p_{k-1} \Rightarrow \frac{p_k}{p_{k-1}} = a+\frac{b}{k}$ $\quad k = 1, 2, 3 \dots$

$p_0 = 1 - \sum_{k=1}^{\infty} p_k$ $\quad$ range $= 0, 1, 2 \dots$

✓

	a	b
$Poi(\lambda)$	0	λ
$Bin(m,q)$	$\frac{-q}{1-q}$	$\frac{(m+1)q}{1-q}$
$NB(r,\beta)$	$\frac{\beta}{1+\beta}$	$\frac{(r-1)\beta}{1+\beta}$

$(a,b,1)$

Same but $k = 2, 3, \dots$ $\quad$ range $= 1, 2, 3 \dots$

Zero-truncated. $p_0^T = 0$ $\quad \{p_k^T\}_{k=0}^{\infty}$, N^T

Zero-modified. $p_0^M \neq 0$ or $p_0^M = p_0 \ (a,b,0)$ $\{p_k^M\}_{k=0}^{\infty}$, N^M

$$\begin{cases} p_k^T = \beta^T p_k = \frac{1}{1-p_0} p_k \quad , \ p_0^T = 0 \\ G_{N^T}(t) = \frac{G_N(t) - p_0}{1-p_0} \end{cases}$$

$$\begin{cases} p_0^M \text{ chosen arbitrarily} \rightarrow \text{also satisfy } p_k^M = (a+\frac{b}{k})p_{k-1}^M \\ p_k^M = \beta^M p_k = \frac{1-p_0^M}{1-p_0} p_k \\ G_{N^M}(t) = \frac{p_0^M - p_0}{1-p_0} + \frac{1-p_0^M}{1-p_0} G_N(t) \\ E((N^M)^k) = \frac{1-p_0^M}{1-p_0} E(N^k) \end{cases}$$

↳ logarithmic (a, b, 1)

$p_k = \frac{p^k}{-k\ln(1-p)}$, $k \geq 1$, $p_0 = 0$

$\frac{p_k}{p_{k-1}} = p + \frac{-p}{k} \Rightarrow a = p$, $b = -p$

$G(t) = \frac{\ln(1-pt)}{\ln(1-p)}$

Mixture ✓

$$P(N=n) = \begin{cases} \sum_{i=1}^{m} P(N=n \mid \theta=\theta_i) P(\theta_i) \\ \int_\theta P(N=n \mid \theta=\theta_i) f(\theta_i)\, d\theta \end{cases}$$

$E(E(N|\theta)) = E(N)$

$Var(N) = E(Var(N|\theta)) + Var(E(N|\theta))$

Compounding

$S = \sum_{i=1}^{N} L_i \Rightarrow g_k$ (pmf)

M_i iid L & $L_1, L_2, \dots$ mutually indep N

primary dist. N , p_k , $G_N(t)$

secondary dist. L , f_k , $G_L(t)$, $M_L(t)$

☆
$$\begin{cases} G_S(t) = E(t^S) = G_N(G_L(t)) \\ M_S(t) = E(e^{tS}) = G_N(M_L(t)) \end{cases}$$

$$\begin{cases} E(S) = E(N)E(L) \\ Var(S) = E(N)Var(L) + E(L)^2 Var(N) \end{cases}$$

$$g_k = P(S=k) = \sum_{n=1}^{\infty} p_n f_k^{*n} \Rightarrow f_k^{*n} = P(L_1 + L_2 \dots L_n = k)$$

$$g_0 = G_N(f_0) = \sum_{n=0}^{\infty} f_0^n p_n \qquad f_k^{*n} = \sum_{i=0}^{k} f_{k-i}^{*(n-1)} f_i$$

Panjer's Recursion:

for $(a,b,0)$ N,

$$g_k = \frac{1}{1-af_0} \sum_{i=1}^{k} \left(a + \frac{bi}{k}\right) f_i g_{k-i}$$

where $g_0 = G_N(f_0)$

for $(a,b,1)$ N

$$g_0 = G_N(f_0)$$

$$g_k = \frac{p_1 - (a+b)p_0}{1-af_0} f_k + \frac{1}{1-af_0} \sum_{i=1}^{k} \left(a + \frac{bi}{k}\right) f_i g_{k-i}$$

Effect of exposure: $\{N_j\}_{j=1}^{a}$

$$N = N_1 + \dots N_a$$

$$G_N(t) = E(t^N) = (G_{N_1}(t))^a$$

Effect of severity: N = total number loss.

$$S = \sum_{i=1}^{N_L} Y_{L,i}, \quad S = \sum_{i=1}^{N_P} Y_{P,i}$$

assume all loss iid, then

$$N_P = \sum_{i=1}^{N_L} I_i \quad \leftarrow \text{Indicator RV.}$$

$$\Rightarrow G_{N_P}(t) = G_{N_L}(G_I(t)) = G_{N_L}(1 - \alpha + \alpha t)$$

MLE: *(complete data)

let $0 < y_1 \le y_2 \cdots \le y_k < \alpha(u-d)$ $\rightarrow k$

and $y_{k+1} = \cdots = y_n = \alpha(u-d)$ $\rightarrow n$

$$L(\theta) = \frac{\left(\prod_{i=1}^{k} f_X(x_i)\right) S_X(u)^{n-k}}{S_X(d)^n} \qquad x_i = \frac{y_i}{\alpha} + d \quad (\text{loss for pmt } y_i)$$

derivative $\Rightarrow l'(\theta) = 0$

Group data estimation. *(grouped data)

pmt intervals: $\underbrace{(c_0, c_1)}_{N_0}$, $\underbrace{(c_1, c_2)}_{N_1}$, $\cdots$ $\underbrace{(c_{n-1}, c_n)}_{N_{n-1}}$, $\underbrace{c_n}_{N_n}$

$0 \rightarrow N_0$ N_n max pmt.

$$L(\theta) = \left(\prod_{i=1}^{N} \left(\frac{S_X(c_{i-1}^*) - S_X(c_i^*)}{S_X(d)}\right)^{N_{i-1}}\right)\left(\frac{S_X(u)}{S_X(d)}\right)^{N_n}$$

$c_i^* =$ "LOSS" for pmt c_i

MLE for frequency. *(other dists.)

$N_P = \sum_{i=1}^{N_L} I_i$ # pmt / loss.

$P(I_i = 1) = P(Y_{L,i} > 0) = \alpha$ review: ✱ $G_{N_P}(t) = G_{N_L}(1 - \alpha + \alpha t)$

estimate N_L by N_P

✱ $L(\theta) = \prod_{k=0}^{\infty} (p_k)^{n_k}$ n_k: # policies w/ k pmts. p_k: $P(N=k)$

grouped:

$$L(\theta) = \left(\prod_{k=0}^{m} (p_k)^{n_k}\right)\left(1 - \sum_{k=0}^{m} p_k\right)^{n_{m+1} + n_{m+2} + n_{m+3} \cdots}$$

(a,b,0) Moments.

☆ $E(N) = \frac{a+b}{1-a}$, $Var(N) = \frac{(a+b)(a+b+1)}{(1-a)^2}$

MLE zero-modified (a,b,0)

$p_0^M = \hat{\alpha} = \frac{n_0}{\sum_{k=1}^{\infty} n_k}$

Chi-square goodness fit.

☆ $Q = \sum_{k=1}^{m} \frac{(n_k - E_k)^2}{E_k} \sim \chi^2_d$, ☆ $E_k = \overset{\text{total}}{n}\, \hat{p}_k$

d.f. = $m-1-r$

H_0: data consistent w/ specific dist

H_A: data not ~

$p = P(\chi^2_d > Q)$

significance level α $\chi^2_{d,\alpha}$

$P(\chi^2_d > \chi^2_{d,\alpha}) = \alpha$

$p < \alpha \Rightarrow$ reject H_0

$p \geq \alpha \Rightarrow$ no evidence against H_0

likelihood ratio test. (two models)

H_0: data follows model 0

H_1: data follows model 1

$Q = 2(l_1 - l_0) \sim \chi^2_d$

$p = P(\chi^2_d > Q)$

Average claim frequency = $n = \dfrac{\#\text{ Incurred claim}}{\text{units of earned exposure}}$.

Average loss severity = $x = \dfrac{\$\text{ Incurred loss}}{\#\text{ Incurred claim}}$.

↳ total Incurred loss = loss paid-to-date + unpaid loss reserve being carried.

Aggregate loss distribution : S or developed from total loss.

Payout pattern : times pmt are made.

Loss cost per unit exposure $= nx$

$= \dfrac{\$\text{ Incurred loss}}{\text{units of earned exposure}}$.

accident year Z

pro : easy , con : data not mature until future.

Policy year Z (effective date)

pro : allow consistent basis premium / loss

cons : 24 month for full data.

Calendar year Z (accounting action)

Loss development :

- Paid age-to-age : pmt so far → pmt later
- age-to-ultimate : today's value → ultimate value.
- Incurred age-to-age : (paid + outstanding) now → ltr.

Case reserve plus method.

total reserve =

1. Case reserve = estimate future pmt.
2. Bulk / Gross IBNR
   - Pure IBNR
   - RBNR
   - future changes in know claims.
   - claim close might reopen.

⇒ total reserve = gross IBNR + all case reserve.

Expected loss ratio method.

estimated ultimate loss = E(loss ratio) × Earned premium.

estimate loss reserve = — loss paid to date.

Total estimate loss reserve = Σ estimate loss reserve.